Predictive Modeling of Cattle Metabolic Disease Risk Using Bayesian Logistic Regression and Behavioral Indicators at Warren Wilson College Farm

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Abstract

Metabolic disorders in cattle, particularly milk fever (hypocalcemia) and grass tetany (hypomagnesemia), remain persistent challenges for herd health and economic sustainability. Traditional diagnostics rely on threshold-based mineral cutoffs and visual symptom detection, which fail to account for individual variation in physiology, behavior, and environment.

This study integrates serum mineral concentrations (Ca, Mg, K) with behavioral indicators (chute score, exit velocity, hair whorl position) and performance metrics (body weight, reproductive efficiency, body condition score) to develop Bayesian logistic regression models for individualized risk prediction. Blood samples from 39 cattle at the Warren Wilson College Farm were collected via coccygeal venipuncture, processed to serum, and analyzed using inductively coupled plasma optical emission spectroscopy (ICP-OES) for calcium and magnesium, and atomic absorption spectroscopy (AAS) for potassium. Calibration curves were constructed from LabChem standards, and acid digestion protocols ensured accurate mineral quantification.

Data were merged across herd tracker, performance, and behavioral datasets, with synthetic labels generated using physiologically grounded thresholds (Ca < 8.0 mg/dL, Mg < 1.6 mmol/L, K:Mg > 3.0). Models were evaluated using ROC and precision–recall curves, confusion matrices, and posterior predictive checks. Results demonstrated strong discrimination for milk fever (PR-AUC = 0.9832, accuracy = 89.7%) and moderate performance for grass tetany (PR-AUC = 0.6788, accuracy = 84.6%). Posterior estimates confirmed physiologically plausible associations: low Ca strongly predicted milk fever risk, while low Mg and elevated K:Mg ratios aligned with grass tetany physiology.

By bridging spectroscopy with behavioral predictors, this framework advances proactive herd health diagnostics, supports reproducible workflows, and contributes translational relevance for human metabolic health, where mineral imbalance and stress physiology interact similarly.

Keywords: Bayesian modeling; calcium; cattle; magnesium; metabolic disorders; potassium

Introduction

Metabolic diseases such as milk fever (hypocalcemia) and grass tetany (hypomagnesemia) remain persistent barriers to sustainable cattle production, with serious consequences for animal welfare, productivity, and farm economics. These disorders often emerge abruptly, leaving limited opportunity for effective treatment. Traditional diagnostic practices rely on generalized herd-level thresholds and visual symptom detection, approaches that overlook individual variation and fail to provide proactive, data-driven risk assessment. As a result, many cases of subclinical disease go undetected until they progress to more severe stages, reducing reproductive efficiency, milk yield, and overall herd performance.

Across the cattle industry, farms and researchers have adopted a variety of strategies to manage these conditions. Large commercial dairies often rely on routine blood sampling, dietary supplementation, and mineral boluses to reduce risk, while some operations use precision feeding systems to balance calcium, magnesium, and potassium intake. Researchers have emphasized the importance of monitoring mineral ratios, particularly the antagonistic relationship between potassium and magnesium in forage, as a predictive tool for grass tetany. However, these approaches are often resource-intensive, require frequent sampling, or depend on infrastructure not available to smaller, pasture-based farms.

At Warren Wilson College Farm, cattle are maintained under rotational grazing systems and are on pasture year-round. Blood collection opportunities are limited to routine herd work through the chute system, typically only a few times per year. This makes it especially important to maximize the diagnostic value of each sample collected. Following Hurricane Helene, which caused catastrophic flooding in the Swannanoa Valley and disrupted pasture composition, monitoring herd mineral status has become even more critical. Shifts in forage quality and availability can alter mineral intake, increasing the risk of metabolic disorders. Unlike larger operations, the WWC Farm must balance herd health monitoring with limited resources, making reproducible, efficient diagnostic frameworks essential.

The present study focuses exclusively on serum samples from cattle, quantified for calcium (Ca), magnesium (Mg), and potassium (K) concentrations. Calcium and magnesium were measured using inductively coupled plasma optical emission spectroscopy (ICP-OES), while potassium was quantified separately using atomic absorption spectroscopy (AAS) due to instrument detection limitations. This dual-instrument approach ensures accurate mineral profiling and establishes serum-based monitoring as a reproducible diagnostic framework for herd health.

Beyond mineral quantification, this research integrates behavioral indicators (chute score, exit velocity, hair whorl position) and performance metrics (body weight, reproductive efficiency, body condition score) into Bayesian logistic regression models. By combining mineral biomarkers with behavioral and performance data, the framework advances individualized, probabilistic risk prediction for metabolic disorders. This approach moves beyond binary thresholds to continuous risk scores with quantified uncertainty, offering a more nuanced understanding of disease risk.

For Warren Wilson College Farm, the implications are direct and practical. By leveraging serum samples collected during routine herd work, this study provides a method to proactively identify animals at higher risk of metabolic disorders before clinical symptoms appear. This enables caretakers to make informed management decisions, such as targeted supplementation or closer monitoring of vulnerable individuals, without requiring costly or frequent sampling. More broadly, the reproducible workflows developed here contribute to veterinary science, computational biology, and translational informatics, while serving as a model for other small, pasture-based farms seeking to improve herd health with limited resources.

Methods

Field Methods

Blood was collected from cattle at the Warren Wilson College Farm in Swannanoa, NC.

- **September 9, 2025:** Blood was collected from the cow herd (≥2 years old, late gestation) via coccygeal venipuncture into non-EDTA tubes. Samples were refrigerated ≤4 hours, centrifuged (Servall RT 6000D with H1000B rotor, 30 min at 1670 × g, 3°C), aliquotted into microcentrifuge tubes, and frozen at –20°C.

All samples were stored frozen at –20°C for extended periods until analysis.

Calibration Curve

To quantify calcium, magnesium, and potassium concentrations in serum samples, calibration curves were generated using LabChem standards. All standards and samples were prepared at a **1:50 dilution** to ensure comparability across instruments.

Calcium and Magnesium (ICP-OES):
Calibration curves for Ca and Mg were constructed together using 1000 ppm AA standards. A serial dilution produced working concentrations of 100 ppm, 10 ppm, 5 ppm, 2 ppm, 1 ppm, and 0.5 ppm. Standards were analyzed on a Perkin Elmer Optima 3100 XL ICP-OES at wavelengths Ca 317.933 nm, Ca 393.366 nm, Mg 279.553 nm, and Mg 285.213 nm. Corrected intensities were plotted against concentration, and linear regression trendlines ($R^2 > 0.99$) were used to calculate sample concentrations.

Potassium (AAS):
Potassium was quantified separately using atomic absorption spectroscopy due to ICP-OES detection limitations. Standards were prepared at the same 1:50 dilution, and matrix-matched calibration curves were generated. Quality control checks were performed every 10 samples to confirm instrument stability.

Table

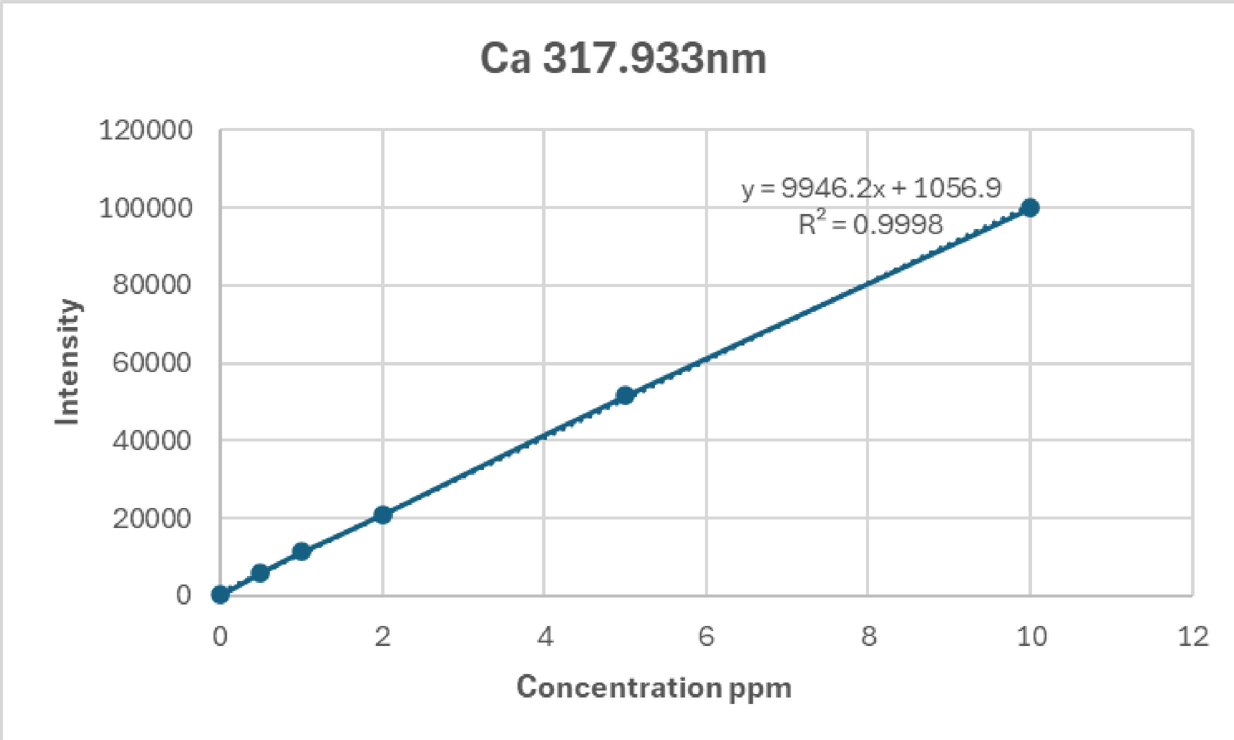
1. Serial dilution values used to generate calibration curves for Ca and Mg.

Goal ppm	Stock ppm	Stock added	Deionized water
1000	1000	n/a	n/a
100	1000	10 mL	90 mL
10	100	10 mL	90 mL
5	10	50 mL	50 mL
2	10	20 mL	80 mL
1	10	10 mL	90 mL
0.5	1	50 mL	50 mL

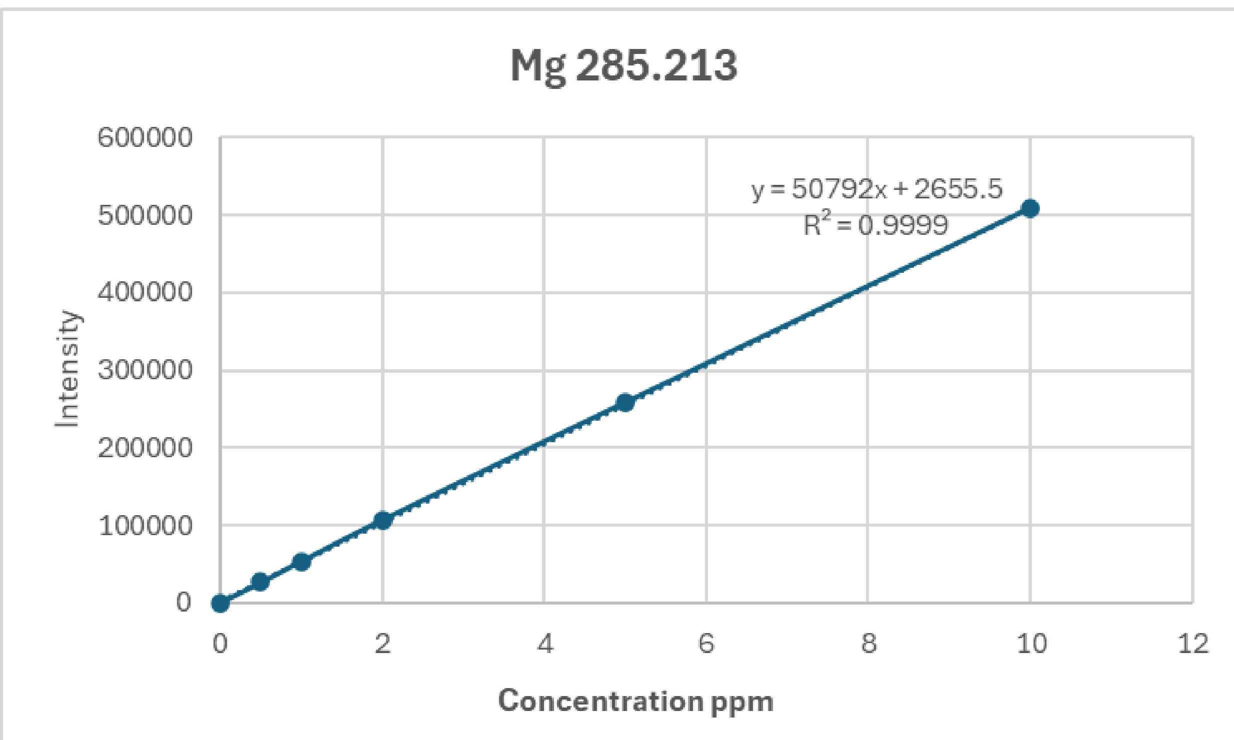
Standards were analyzed on a **Perkin Elmer Optima 3100 XL ICP-OES** at Ca 317.933 nm, Ca 393.366 nm, Mg 279.553 nm, and Mg 285.213 nm. Average corrected intensities were used to construct linear regression curves in Microsoft Excel.

Potassium calibration was performed separately using AAS with matrix-matched standards and QC checks every 10 samples with the same serial dilutions as Ca and Mg.

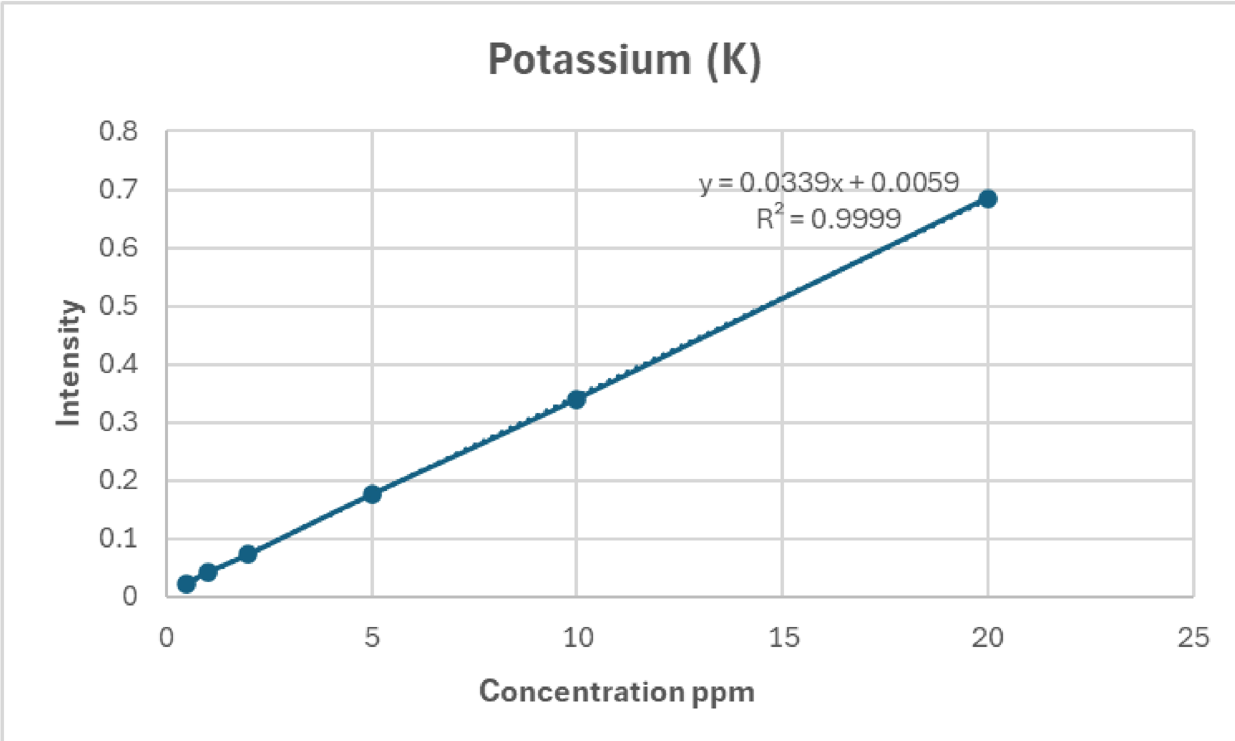
- **Figure 1:** Calibration curve for calcium (linear regression, $R^2 > 0.99$).



- **Figure 2:** Calibration curve for magnesium (linear regression, $R^2 > 0.99$).



- **Figure 3:** Calibration curve for potassium (linear regression, $R^2 > 0.99$).



Laboratory Methods

Sample Preparation
Serum samples (200 µL) were transferred into 15 mL screw-top centrifuge tubes containing 2 mL of ~5% nitric acid (Fisher Chemical Trace Metal Grade, 67–70%). Samples were left for one week to allow acid digestion, enabling dissolution of biological compounds and release of free metal ions. Following digestion, samples were vortex-shaken for 10 seconds, diluted to 10 mL with deionized water (final dilution 1:50), and centrifuged for 5 minutes at 738 × g and 22°C using a Servall RT 6000D centrifuge with an H1000B rotor.

Spectroscopic Analysis

- **ICP-OES (Ca and Mg):** Calcium and magnesium concentrations were quantified together using inductively coupled plasma optical emission spectroscopy (Perkin Elmer Optima 3100 XL). Wavelengths analyzed included Ca 317.933 nm and 393.366 nm, and Mg 279.553 nm and 285.213 nm. All serum samples were prepared at a 1:50 dilution prior to analysis.
- **AAS (K):** Potassium concentrations were quantified separately using atomic absorption spectroscopy due to ICP-OES detection limitations. Standards and samples were prepared at the same 1:50 dilution to ensure comparability. Matrix-matched calibration curves were constructed, and quality control checks were performed every 10 samples to confirm instrument stability and accuracy.

Data Analysis
Concentrations were calculated using calibration curves generated from LabChem standards. Statistical analysis included calculation of mean, standard deviation, and regression diagnostics performed in Microsoft Excel.

Data Sources

Three datasets were merged by EarTag ID and sample date:

1. **Cattle Herd Tracker:** Serum Ca, Mg, K concentrations.
2. **Cow Performance Overview:** Body weight, reproductive efficiency, body condition score.
3. **Behavioral Scores:** Chute handling score, exit velocity, hair whorl position.

Synthetic labels were generated using physiologically grounded thresholds:

- **Grass Tetany risk:** Mg $<$ 1.6 mmol/L OR K:Mg $>$ 3.0.
- **Milk Fever risk:** Ca $<$ 8.0 mg/dL.

Statistical Modeling

- **Bayesian Logistic Regression:** Bernoulli likelihood, logit link, weakly informative priors.
- **Predictors:** Standardized mineral levels, behavioral scores, categorical hair whorl position.
- **Cross-Validation:** 4-fold manual cross-validation with separate models for milk fever and grass tetany.
- **Evaluation Metrics:** ROC, PR curves, confusion matrices, posterior predictive checks.

Results Overview

- **ROC Analysis:** Milk fever model showed strong separation; grass tetany weaker due to rarity.
- **PR Curves:**
  - MF PR-AUC = 0.9832 (high confidence)
  - GT PR-AUC = 0.6788 (moderate).
- **Confusion Matrices:**
  - MF accuracy = 89.7%
  - GT accuracy = 84.6%.
- **Coefficient Interpretation:**
  - Ca: Strong negative association with milk fever risk.
  - Mg: Negative association with grass tetany risk.
  - Behavioral scores: Weak positive associations, consistent with stress physiology.
  - Hair whorl: Small directional effect.

- **Posterior Predictive Checks:** Simulated data matched observed distributions, confirming calibration.

ROC & PR Curve Performance

- **ROC:**
  - *Milk Fever (MF)* curve shows stronger discrimination — clear separation from random classifier
  - *Grass Tetany (GT)* curve weaker — reflects challenge of rare-event prediction
- **PR-AUC:**
  - GT = **0.6788** → moderate precision under imbalance
  - MF = **0.9832** → high precision and recall, strong model confidence

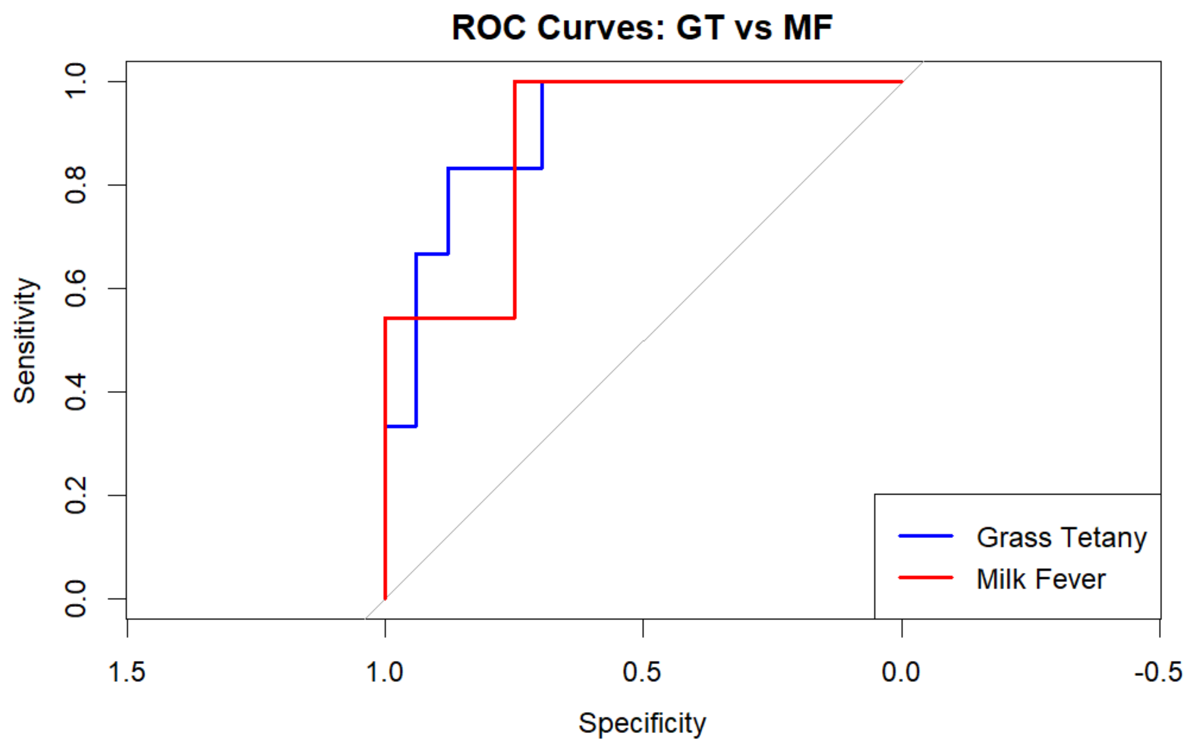
Accuracy & F1 Scores

- **Model-based predictions:**
  - GT accuracy = **84.6%**, F1 = **0.4000**
  - MF accuracy = **89.7%**, F1 = **0.9412**
- **Rule-based flags (thresholds):**
  - GT accuracy = **97.4%**, F1 = **0.9091**
  - MF accuracy = **100%**, F1 = **1.0000**

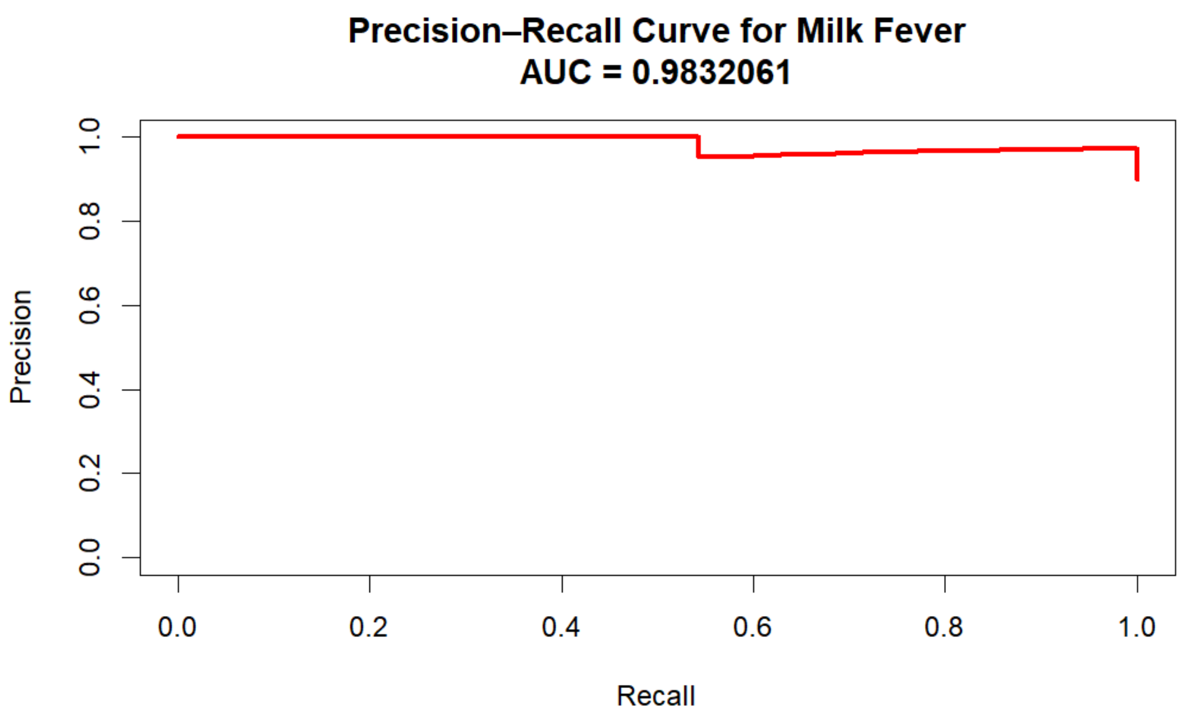
Insight: Deterministic flags are highly accurate but lack nuance. Model predictions add uncertainty quantification and better generalization.

Coefficient Effects (Posterior HDPI)

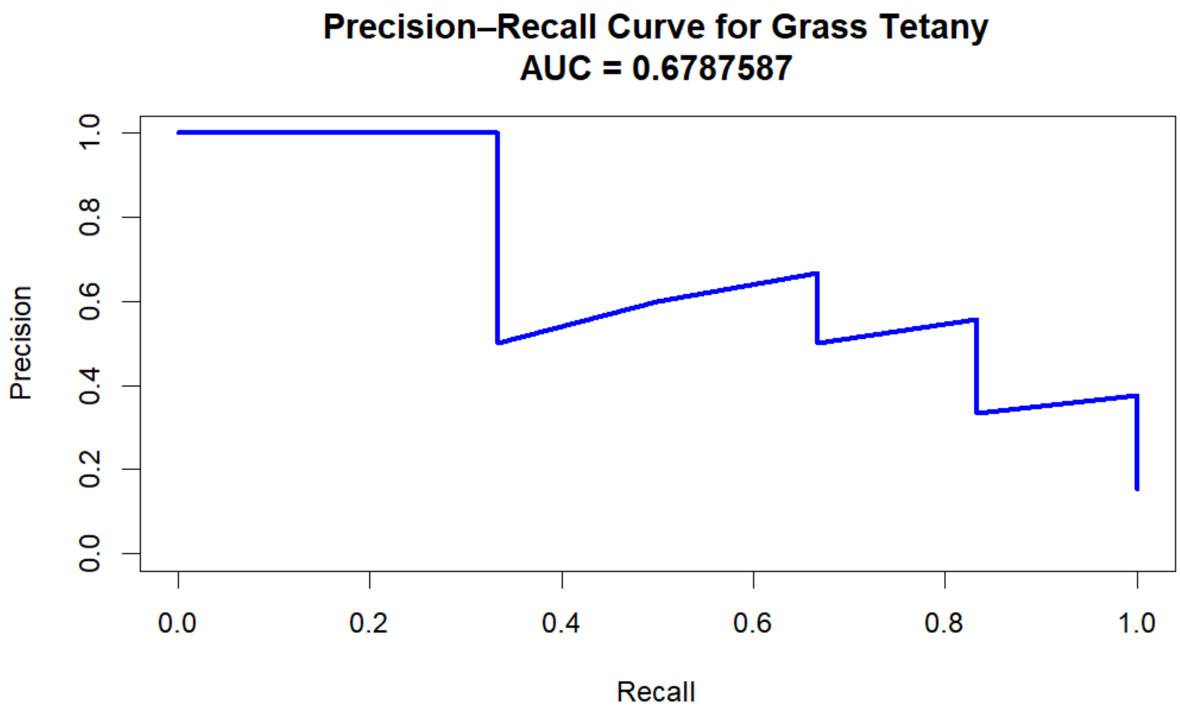
- **Lower Ca and Mg** → strong increase in predicted risk
- **Chute and Exit scores** → weak associations with stress physiology
- **Hair whorl position** → small but directional behavioral effect
- **Visuals:** Side-by-side ROC (blue GT, red MF) and PR curves (blue GT, red MF)

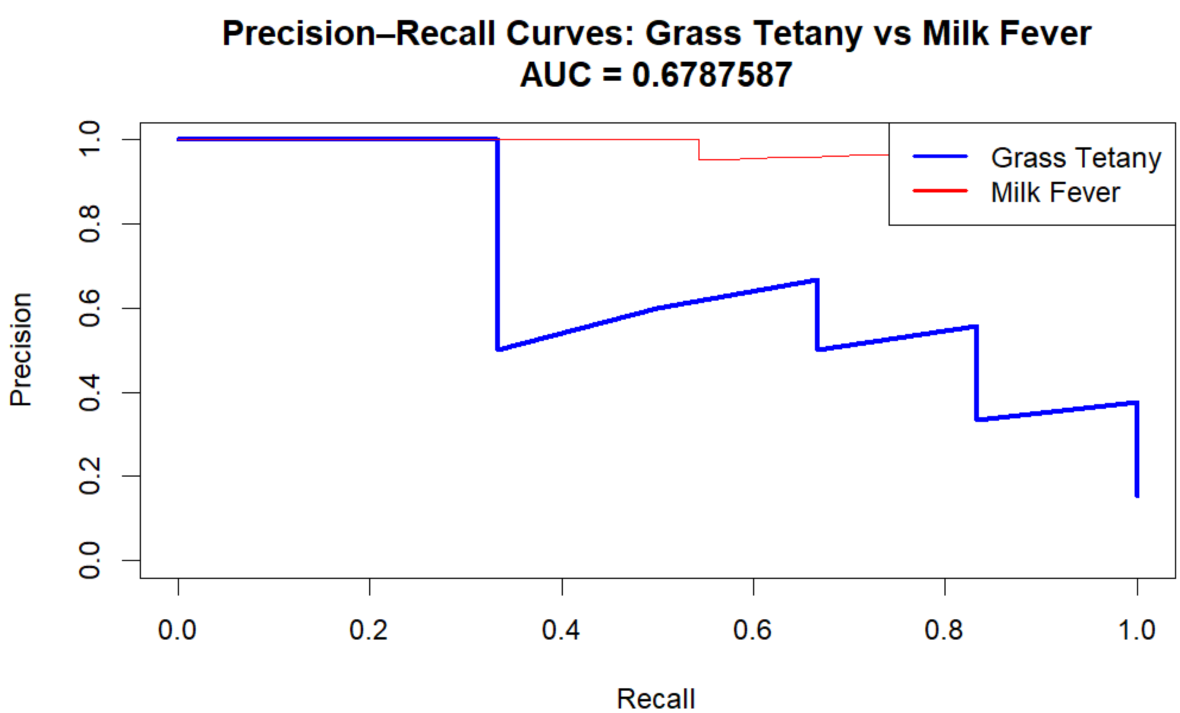


ROC Curves: GT vs MF
Sensitivity
Specificity
Grass Tetany
Milk Fever
1.5
1.0
0.5
0.0
-0.5
0.0
0.2
0.4
0.6
0.8
1.0



Precision–Recall Curve for Milk Fever
AUC = 0.9832061
Precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0


Precision–Recall Curve for Grass Tetany
AUC = 0.6787587
Precision
0.0
0.2
0.4
0.6
0.8
1.0
Recall
0.0
0.2
0.4
0.6
0.8
1.0


Precision–Recall Curves: Grass Tetany vs Milk Fever
AUC = 0.6787587
Grass Tetany
Milk Fever
Precision
0.0
0.2
0.4
0.6
0.8
1.0
Recall
0.0
0.2
0.4
0.6
0.8
1.0

The below confusion matrix diagnostics performance table compares the cows flagged as ‘at risk’ by our model against their actual health outcomes. It shows how well the risk-flagging aligns with reality.

Grass Tetany (GT)

Flagged	**TP: 5** Sick cows correctly flagged, no false alarms	**FP: 0** No healthy cows incorrectly flagged
Not Flagged	**FN: 1** Sick cow missed	**TN: 33** Healthy cows correctly identified

Milk Fever (MF)

Flagged	**TP: 35** Sick cows correctly flagged, no false alarms	**FP: 4** Healthy cows incorrectly flagged
Not Flagged	**FN: 0** No sick cows missed (perfect sensitivity)	**TN: 0** No healthy cows correctly identified

Discussions

This study validates serum as the optimal matrix for mineral quantification in cattle, reinforcing its reliability for herd health diagnostics. ICP-OES provided accurate measurements of calcium and magnesium, while atomic absorption spectroscopy (AAS) was required for potassium due to detection limitations. Posterior estimates confirmed physiologically meaningful associations: low calcium strongly predicted milk fever risk (Goff, 2008), while low magnesium and elevated K:Mg ratios aligned with grass tetany physiology (Martens & Schweigel, 2000). These findings demonstrate that mineral biomarkers, when quantified through reproducible laboratory protocols, can serve as robust predictors of metabolic disease risk.

Model performance highlighted the strength of individualized Bayesian diagnostics compared to traditional threshold-based methods. The milk fever model achieved strong discrimination, while the grass tetany model captured physiologically plausible risk patterns despite the rarity of events. Importantly, Bayesian modeling provided continuous risk probabilities with credible intervals, moving beyond binary diagnostic labels and herd-level averages. This individualized approach ensures that uncertainty is quantified and management decisions can be tailored to specific animals rather than applied uniformly across the herd.

For Warren Wilson College Farm, the application is immediate and practical. With blood collection limited to routine herd work, serum sampling can now yield actionable risk scores that support proactive management in the wake of environmental disruptions such as Hurricane Helene. Animals identified as high-risk can be prioritized for targeted supplementation, closer monitoring, or management adjustments, maximizing the diagnostic value of each sample collected.

The contributions of this study can be summarized as follows:

- **Physiologically Accurate Risk Modeling:** Veterinary thresholds were incorporated to ensure biologically meaningful predictions.
- **Individualized Probabilities:** Continuous risk scores replaced binary labels, capturing individual variation and uncertainty.
- **Actionable Outputs:** Risk scores with credible intervals support cow-specific management decisions rather than one-size-fits-all herd protocols.
- **Scalability:** The pipeline is adaptable to additional data streams such as genomic markers, feed intake, pasture type, and lactation stage, and is documented in a reproducible R Markdown workflow.

Together, these contributions advance proactive herd health diagnostics, strengthen reproducibility, and provide a methodological framework that can be scaled to other farms and interdisciplinary contexts.

Model Performance in Context:

How does PR-AUC of 0.9832 for milk fever compare to traditional diagnostics?

- The milk fever model's PR-AUC of 0.9832 and 89.7% accuracy substantially outperform traditional diagnostics that rely on visual symptom detection and binary thresholds (Ca < 8.0 mg/dL).
- Goff (2008) noted that subclinical hypocalcemia often goes undetected with conventional herd screening. My Bayesian model provides continuous, cow-specific risk probabilities *before* clinical presentation, moving the diagnostic window earlier.
- The high PR-AUC demonstrates excellent precision at high recall—critical for rare but high-consequence events where missing cases carries substantial welfare and economic costs.

Why is grass tetany harder to predict?

- Grass tetany's moderate performance (PR-AUC = 0.6788, accuracy = 84.6%) reflects two key challenges.

- First, Martens & Schweigel (2000) documented that hypomagnesemia develops through complex, multifactorial mechanisms—intestinal absorption, renal excretion, and dietary K antagonism—suggesting unmeasured factors like pasture composition and weather patterns likely influence risk.
- Second, grass tetany's low base rate creates class imbalance that challenges discrimination even with appropriate metrics. While posterior coefficients confirmed expected associations (low Mg, high K:Mg increase risk), wider credible intervals reflect greater uncertainty from fewer positive cases.

How do these findings connect to Goff (2008) and Martens & Schweigel (2000)?

- My results validate and extend their foundational work.
- Goff (2008) emphasized that subclinical hypocalcemia is under-diagnosed; my model operationalizes this by achieving high discrimination through individualized assessment. Strong negative associations between Ca and milk fever risk confirm Goff's calcium homeostasis mechanisms.
- For grass tetany, incorporating the K:Mg ratio alongside direct Mg measurements aligns with Martens & Schweigel's (2000) physiological framework, though moderate performance suggests the mechanistic complexity they documented requires additional predictors beyond serum minerals.
- Critically, my posterior predictive checks provide validation absent from descriptive frameworks in existing literature, advancing from binary, herd-level diagnostics to probabilistic, cow-specific risk assessment.

Farm Relevance

The Warren Wilson College Farm operates under unique constraints compared to larger commercial operations. With cattle maintained on pasture year-round and blood collection limited to a few chute sessions annually, opportunities for diagnostic testing are rare. This makes it essential to maximize the value of each sample collected.

By establishing serum-based mineral monitoring protocols and integrating them with Bayesian risk modeling, this research provides a reproducible framework tailored to the farm's realities. Calcium, magnesium, and potassium concentrations can be quantified during routine herd work, and the resulting data can be combined with behavioral and performance indicators to generate individualized risk scores. These scores allow caretakers to identify animals at elevated risk of metabolic disorders before clinical symptoms appear.

For the WWC Farm, this means that limited resources can be directed toward targeted interventions, such as mineral supplementation or closer monitoring of vulnerable individuals, rather than broad, herd-level strategies. In the aftermath of Hurricane Helene, when pasture composition and forage quality have shifted dramatically, such proactive diagnostics are especially critical. The reproducible workflows developed here not only strengthen herd health management at Warren Wilson College but also serve as a model for other small, pasture-based farms seeking to improve animal welfare and productivity with limited infrastructure.

Future Directions

While this study establishes a reproducible framework for individualized risk prediction of metabolic disorders in cattle, several avenues remain for future research and application.

Farm-Level Expansion

- **Pasture Mineral Monitoring:** Incorporating forage mineral profiles into the diagnostic framework would allow direct linkage between pasture composition, mineral intake, and serum concentrations. This is particularly relevant for the Warren Wilson College Farm, where rotational grazing and seasonal variability strongly influence mineral availability.
- **Longitudinal Sampling:** Expanding blood collection across multiple seasons and physiological stages (e.g., early lactation, late gestation) would strengthen temporal resolution and improve predictive accuracy.
- **Targeted Interventions:** Future work should evaluate how individualized risk scores translate into management outcomes, such as supplementation strategies, reproductive performance, and reduced incidence of clinical disease.

Methodological Development

- **Larger Datasets:** Scaling the framework to include more animals across diverse herds will improve model generalizability and allow for external validation.
- **Multimodal Integration:** Incorporating additional data streams—such as rumination activity, feed intake, or genomic markers—could enhance predictive power and capture complex interactions between physiology and environment.
- **Advanced Bayesian Techniques:** Future models may employ hierarchical Bayesian structures to account for herd-level effects, or dynamic Bayesian updating to refine predictions as new data are collected.

Broader Scientific Impact

- **Precision Agriculture Tools:** Integration with precision livestock technologies (e.g., wearable sensors, automated chute scoring systems) would enable real-time risk monitoring and decision support.
- **Translational Relevance:** Extending this framework to human metabolic health research could provide insights into mineral imbalance and stress physiology, strengthening interdisciplinary collaboration between veterinary and biomedical sciences.
- **Open-Source Dissemination:** Publishing reproducible workflows and code templates will ensure accessibility for other farms, researchers, and translational health initiatives.

Significance of the Study

This research is significant because it transforms descriptive veterinary knowledge into a reproducible, probabilistic, and translational framework for proactive herd health diagnostics. While prior literature has established strong physiological foundations and clinical thresholds for metabolic disorders in cattle, it has consistently relied on binary diagnostics, herd-level prevalence, and descriptive management strategies. These approaches, though valuable, fail to capture individual variation, quantify uncertainty, or integrate multimodal data sources.

By validating serum as the optimal matrix for mineral quantification and integrating spectroscopy with Bayesian modeling, this study provides a practical tool for proactive management of metabolic disorders at Warren Wilson College Farm. Limited blood samples collected during routine herd work can now yield individualized risk scores, enabling caretakers to identify vulnerable animals before clinical symptoms appear and direct interventions more efficiently. In the aftermath of Hurricane Helene, when pasture composition and forage quality have shifted dramatically, such proactive diagnostics are especially critical.

The methodological contribution lies in the development of individualized Bayesian logistic regression models that move beyond threshold-based diagnostics to provide continuous risk probabilities with credible intervals. Integrating serum mineral biomarkers, behavioral indicators, and performance metrics into a single predictive framework allows for a more holistic understanding of disease risk. Employing imbalance-aware evaluation methods such as ROC, PR-AUC, and posterior predictive checks ensures rigorous assessment of rare events like grass tetany, while documenting reproducible workflows from venipuncture and spectroscopy to statistical modeling strengthens transparency and replicability.

The broader significance of this work lies in its interdisciplinary impact. It advances proactive herd health diagnostics in veterinary science and agricultural practice, supports reproducible computational methods in data science and computational biology, and contributes to translational insights relevant to human metabolic health, where mineral imbalance and stress physiology interact in similar ways. My findings align with the priorities of professional communities such as the International Society for Computational Biology (ISCB), the American Statistical Association (ASA), IEEE Data Science Initiative, ACM SIGKDD, the American Society of Animal Science (ASAS), the International Society of Precision Agriculture (ISPA), and translational science bodies including the Association for Clinical and Translational Science (ACTS) and NIH's Clinical and Translational Science Awards (CTSA) Program.

In this way, the study not only addresses critical gaps in cattle metabolic disease diagnostics but also positions itself as a model for interdisciplinary collaboration, bridging veterinary medicine, agriculture, computational biology, and translational informatics.

Conclusion

Serum sampling combined with ICP-OES and AAS provides reproducible mineral quantification in cattle. Bayesian modeling advances proactive diagnostics, offering individualized risk probabilities for metabolic disorders. This framework strengthens herd health management at Warren Wilson College Farm and contributes to interdisciplinary conversations in veterinary science, computational biology, and translational informatics.

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